

scripts/09_build_presentation.py

382 lines

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1 """
2 09_build_presentation.py
3 =====
4 Builds the submission deck from the hackathon's OWN template
5 (`report/HACKATHON_TEMPLATE.pptx`, the SIH Idea-Submission format).
6
7 Template rules that are respected here:
8 * "Kindly keep the maximum slides limit up to six (6) (including the title
9 slide)" -> exactly 6 slides
10 * "Try to avoid paragraphs and post your idea in points / diagrams /
11 Infographics" -> bullets and figures only, no prose blocks
12 * the template's own title placeholders, footers and slide numbers are kept
13
14 The template's five content sections are mapped onto the content the problem
15 statement demands:
16
17 Template section Problem-statement content
18 -----
19 2. Idea / Proposed Solution Process Understanding & Model
20 3. Technical Approach Control Strategy
21 4. Feasibility & Viability Safety performance + robustness evidence
22 5. Artifacts Results, scenario plots, dashboard snaps
23 6. Research and References Lessons learned + references
24
25 Every number written into the deck is READ FROM THE GENERATED DATA, never
26 hard-coded, so the deck cannot drift from the results.
27 """
28
29 from __future__ import annotations
30
31 import json
32 import sys
33 from pathlib import Path
34
35 ROOT = Path(__file__).resolve().parents[1]
36 sys.path.insert(0, str(ROOT / "src"))
37
38 try:
39     from pptx import Presentation
40     from pptx.dml.color import RGBColor
41     from pptx.util import Emu, Inches, Pt
42 except ImportError: # optional dependency - never break the core pipeline
43     print(" python-pptx not installed - skipping deck build.")
44     print(" (pip install python-pptx) see requirements-optional.txt")
45     raise SystemExit(0)
46
47 from config import CONTROLLER, DATA_DIR, LIMITS, REPORT_DIR, STEP_SEQUENCE_ID
48
49 TEMPLATE = REPORT_DIR / "HACKATHON_TEMPLATE.pptx"
50 OUT = ROOT / "submission" / "Autonomous_Choke_Controller_Presentation.pptx"
51 FIG = ROOT / "figures"
52
53 INK = RGBColor(0x0F, 0x17, 0x2A)
54 DIM = RGBColor(0x52, 0x5F, 0x73)
55 BLUE = RGBColor(0x1D, 0x4F, 0xD8)
56
57 GREEN = RGBColor(0x11, 0x7A, 0x3A)
58 RED = RGBColor(0xB9, 0x1C, 0x1C)
59
60 # ----- helpers
61 def load_results():
62     s = json.loads((DATA_DIR / "scenario_summary.json").read_text())
63     r = json.loads((DATA_DIR / "robustness_summary.json").read_text())
64     m = json.loads((DATA_DIR / "model_validation_metrics.json").read_text())
65     fm = json.loads((DATA_DIR / "fitted_model.json").read_text())
66     return s, r, m, fm
67
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68
69 def clear_body_shapes(slide, keep_title=True):
70 """Remove the template's placeholder body text boxes, keeping the title,
71 footer and slide-number placeholders intact."""
72 for shp in list(slide.shapes):
73 name = shp.name or ""
74 if "Footer" in name or "Slide Number" in name:
75 continue
76 if keep_title and shp == slide.shapes.title:
77 continue
78 if shp.has_text_frame and shp != slide.shapes.title:
79 shp._element.getparent().remove(shp._element)
80
81
82 def set_title(slide, text, size=30):
83 t = slide.shapes.title
84 if t is None:
85 return
86 t.text_frame.text = text
87 for p in t.text_frame.paragraphs:
88 for run in p.runs:
89 run.font.size = Pt(size)
90 run.font.bold = True
91 run.font.color.rgb = INK
92
93
94 def add_bullets(slide, left, top, width, height, items, size=12.5, gap=5):
95 """items: list of (text, level, bold, colour|None)"""
96 box = slide.shapes.add_textbox(Inches(left), Inches(top), Inches(width), Inches(height))
97 tf = box.text_frame
98 tf.word_wrap = True
99 first = True
100 for text, level, bold, colour in items:
101 p = tf.paragraphs[0] if first else tf.add_paragraph()
102 first = False
103 p.level = level
104 p.space_after = Pt(gap)
105 run = p.add_run()
106 run.text = text
107 run.font.size = Pt(size)
108 run.font.bold = bold
109 run.font.color.rgb = colour or (INK if bold else DIM)
110 return box
111
112
113 def add_pic(slide, path, left, top, max_w, max_h, center_in_box=True):
114 """Insert a picture scaled to FIT inside a max_w x max_h box, preserving
115 aspect ratio. Sizing by width alone silently pushes tall figures off the
116 bottom of the slide, so both dimensions are always bounded here."""
117 path = Path(path)
118 if not path.exists():
119 print(f" (missing figure: {path.name})")
120 return None
121 from PIL import Image
122 iw, ih = Image.open(path).size
123 scale = min(max_w / (iw / 96), max_h / (ih / 96))
124 w, h = (iw / 96) * scale, (ih / 96) * scale
125 x = left + (max_w - w) / 2 if center_in_box else left
126 y = top + (max_h - h) / 2 if center_in_box else top
127 return slide.shapes.add_picture(str(path), Inches(x), Inches(y),
128 width=Inches(w), height=Inches(h))
129
130
131 def add_band(slide, left, top, width, height, text, fill, size=11):
132 """A small coloured KPI band."""
133 from pptx.enum.shapes import MSO_SHAPE
134 shp = slide.shapes.add_shape(MSO_SHAPE.ROUNDED_RECTANGLE,
135 Inches(left), Inches(top), Inches(width), Inches(height))
136 shp.fill.solid()
137 shp.fill.fore_color.rgb = fill

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138 shp.line.fill.background()
139 shp.shadow.inherit = False
140 tf = shp.text_frame
141 tf.word_wrap = True
142 tf.margin_left = tf.margin_right = Emu(45720)
143 tf.text = text
144 for p in tf.paragraphs:
145     for run in p.runs:
146         run.font.size = Pt(size)
147         run.font.bold = True
148         run.font.color.rgb = RGBColor(0xFF, 0xFF, 0xFF)
149     return shp
150
151
152 # ----- build
153 def main():
154     if not TEMPLATE.exists():
155         print(f"Template not found at {TEMPLATE} - skipping deck build.")
156     return
157
158 s, r, m, fm = load_results()
159 meta = r["_meta"]
160 q_max = meta["q_max_safe_ground_truth"]
161 pct_of_max = 100 * r["C"]["final_Q_mean"] / q_max
162 n_runs = meta["n_runs_per_scenario"] * 3
163
164 prs = Presentation(str(TEMPLATE))
165
166 # Drop the template's instructions slide (slide 1) -> leaves exactly 6.
167 xml_slides = prs.slides._sldIdLst
168 slides = list(xml_slides)
169 xml_slides.remove(slides[0])
170
171 S = prs.slides
172 assert len(S) == 6, f"expected 6 slides after removing instructions, got {len(S)}"
173
174 # ----- 1: Title
175 sl = S[0]
176 clear_body_shapes(sl, keep_title=False)
177 add_bullets(sl, 0.7, 1.5, 11.9, 1.4, [
178     ("Autonomous Production Choke Controller", 0, True, INK),
179 ], size=34)
180 add_bullets(sl, 0.7, 2.6, 11.9, 0.6, [
181     ("for a Single Naturally Flowing Oil Well", 0, False, DIM),
182 ], size=18)
183 add_bullets(sl, 0.7, 3.5, 11.9, 2.4, [
184     ("Brute-force Model Predictive Control · control interval Ts = 1 hour", 0, False, DIM),
185     ("Maximises safe production; refuses targets that cannot be met safely", 0, False, DIM),
186     ("", 0, False, DIM),
187     (f"0 constraint violations across {n_runs} closed-loop runs | "
188      f"targets tracked to <1 % | {pct_of_max:.1f} % of max safe rate when target infeasible",
189     0, True, GREEN),
190 ], size=14)
191
192 # ----- 2: Process Understanding & Model
193 sl = S[1]
194 clear_body_shapes(sl)
195 set_title(sl, "PROCESS UNDERSTANDING & MODEL", 26)
196 add_bullets(sl, 0.55, 1.30, 6.0, 5.3, [
197     ("Step-test results (our own experiments)", 0, True, BLUE),
198     (f"{len(STEP_SEQUENCE_ID)} choke steps, up and down, 5–20 % magnitude, each held 8 h "
199      "(≈4× slowest τ) to reach true steady state", 1, False, None),
200     ("Opening the choke raises Q and FLP, lowers WHP and BHP – one input, "
201      "four strongly coupled outputs", 1, False, None),
202     ("Gain dQ/du collapses 3.91 → 0.15 bbl/hr per % across the range (~26×) "
203      "→ the process is strongly nonlinear", 1, False, None),
204     ("Provided reference dataset used for illustration only – never for fitting, "
205      "per the problem statement", 1, False, None),
206     ("", 0, False, None),
207     ("Model assumptions", 0, True, BLUE),

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208 ("Linear IPR (constant productivity index); turbulent tubing friction  $\propto Q^2$ ; "
209 "choke orifice  $Q = C_v \cdot f(u) \cdot \sqrt{(WHP-FLP)}$ ", 1, False, None),
210 ("One first-order lag per output; dead time negligible ( $\theta \leq 0.09$  h)", 1, False, None),
211 ("WHT and AP monitored and logged, but not active constraints", 1, False, None),
212 ("", 0, False, None),
213 ("Dynamic model developed", 0, True, BLUE),
214 (" $y(k+1) = y(k) + \alpha \cdot (y_{ss}(u) + d - y(k))$ ,  $\alpha = 1 - e^{-(Ts/\tau)}$ ", 1, True, INK),
215 ("Nonlinearity captured as a piecewise-linear steady-state curve  $y_{ss}(u)$ ; "
216 "dynamics stay linear  $\rightarrow$  simple and explainable", 1, False, None),
217 (f"Fitted  $\tau$ :  $Q \{fm['outputs']['Q']['\tau']:.2f\} h \cdot WHP \{fm['outputs']['WHP']['\tau']:.2f\} h \cdot "$ 
218 f"FLP  $\{fm['outputs']['FLP']['\tau']:.2f\} h \cdot BHP \{fm['outputs']['BHP']['\tau']:.2f\} h$ ", 1, False, None),
219 (f"Validated on HELD-OUT steps: NRMSE  $\{m['Q']['nrmse_pct']:.2f\} \% (Q)$ , "
220 f" $\{m['BHP']['nrmse_pct']:.2f\} \% (BHP)$ ", 1, True, GREEN),
221 ], size=11)
222 add_pic(sl, FIG / "step_gain_curve.png", 6.80, 1.22, 6.15, 2.15)
223 add_pic(sl, FIG / "model_validation.png", 6.80, 3.45, 6.15, 3.30)
224
225 # ----- 3: Technical Approach / Control Strategy
226 sl = S[2]
227 clear_body_shapes(sl)
228 set_title(sl, "TECHNICAL APPROACH – CONTROL STRATEGY", 24)
229 add_bullets(sl, 0.55, 1.25, 6.6, 5.4, [
230 ("Prediction methodology", 0, True, BLUE),
231 (f"Enumerate ALL candidate moves within the ramp limit:  $u \pm \{CONTROLLER.max\_move:.0f\} \%$  "
232 f"on a  $\{CONTROLLER.candidate\_step\} \%$  grid  $\rightarrow$  21 candidates", 1, False, None),
233 (f"Roll each forward  $\{CONTROLLER.horizon\} h$  through the identified model "
234 "(move-blocked: apply, then hold)", 1, False, None),
235 (f"Horizon must outrun the slowest lag:  $BHP \tau = \{fm['outputs']['BHP']['\tau']:.2f\} h$ , "
236 f"so  $\{CONTROLLER.horizon\} h \approx 5.7 \tau$ ", 1, False, None),
237 ("Every prediction restarts from the current measurement  $\rightarrow$  model error never compounds", 1, False, None),
238 ("Measurement filter + bias correction remove hunting and steady-state offset", 1, False, None),
239 ("", 0, False, None),
240 ("Choke move selection logic", 0, True, BLUE),
241 (" $J = (Q_{pred\_end} - Q_{target})^2 + 0.05 \cdot (\Delta u)^2 \rightarrow$  closest to target, penalise large moves", 1, True, INK),
242 ("Deterministic tie-breaks: lowest cost  $\rightarrow$  smallest  $|\Delta u| \rightarrow$  lower choke (conservative)", 1, False, None),
243 ("Deadband: move only if it buys  $\geq 2$  (bbl/hr) $^2$  improvement  $\rightarrow$  zero valve chatter", 1, False, None),
244 ("", 0, False, None),
245 ("Constraint handling", 0, True, BLUE),
246 (f"Reject any candidate whose predicted WHP/FLP/BHP breaches limits at ANY point in the horizon",
247 1, False, None),
248 (f"Safety margins inside each hard limit ( $\{CONTROLLER.margin\_whp:.0f\}/\{CONTROLLER.margin\_flp:.0f\}/"$ 
249 f" $\{CONTROLLER.margin\_bhp:.0f\} \psi$ ) cover sensor noise + model error + filter lag", 1, False, None),
250 ("If ALL candidates unsafe  $\rightarrow$  pick the least-bad (smallest predicted violation), never lurch", 1, False, None),
251 ("Infeasible target needs no special case: 'open further' is filtered out every "
252 "step  $\rightarrow$  the controller pins itself at the safe boundary", 1, False, None),
253 ], size=9.8, gap=4)
254 add_pic(sl, FIG / "dashboard" / "dash_reasoning.png", 7.35, 1.30, 5.55, 5.15)
255 add_bullets(sl, 7.45, 6.62, 5.4, 0.4, [
256 ("Live MPC reasoning: every candidate, why each was rejected", 0, False, DIM),
257 ], size=9)
258
259 # ----- 4: Feasibility & Viability
260 sl = S[3]
261 clear_body_shapes(sl)
262 set_title(sl, "FEASIBILITY AND VIABILITY", 26)
263 kpis = [
264 (f" $\{n\_runs\}$  \nclosed-loop runs", BLUE),
265 (" $\{n\_constraint\_violations\}$ ", GREEN),
266 (f" $\{pct\_of\_max:.1f\} \%$  \nof max safe rate", GREEN),
267 (" $\{n\_tests\_passing\}$ ", BLUE),
268 ]
269 for i, (txt, col) in enumerate(kpis):
270 add_band(sl, 0.55 + i * 3.12, 1.28, 2.92, 0.95, txt, col, size=12)
271
272 add_bullets(sl, 0.55, 2.45, 6.2, 4.4, [
273 ("Feasibility – proven, not asserted", 0, True, BLUE),
274 (f"Monte-Carlo:  $\{meta['n\_runs\_per\_scenario']\}$  independent noise seeds  $\times$  3 scenarios; "
275 "envelope checked on every sample", 1, False, None),

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276 (f"Worst margin to any limit: A +{r['A']['worst_margin_psi']:.0f} psi . "
277 f"B +{r['B']['worst_margin_psi']:.0f} psi . C +{r['C']['worst_margin_psi']:.1f} psi", 1, False, None),
278 ("Runs end-to-end in ~2 min; no GPU, no internet, no solver library", 1, False, None),
279 ("", 0, False, None),
280 ("Challenges found – and fixed", 0, True, RED),
281 ("Horizon shorter than the slowest lag → BHP kept falling after the horizon "
282 "(11/100 runs breached)", 1, False, None),
283 ("Coarse step-test knots → chord across a convex curve over-estimated BHP by "
284 "8 psi (unsafe direction)", 1, False, None),
285 ("Both passed single-run testing; only the Monte-Carlo exposed them", 1, True, INK),
286 ("", 0, False, None),
287 ("Risk strategy", 0, True, BLUE),
288 ("Margins sized for noise + model error + filter lag, and re-verified over 300 runs", 1, False, None),
289 ("Cost of safety quantified, not hidden: {meta['margin_cost_bblhr']:.1f} bbl/hr "
290 f"({100*meta['margin_cost_bblhr']/q_max:.1f} %) below the theoretical max", 1, False, None),
291 ("Controller reads ONLY step() outputs → official simulator drops in with zero "
292 "changes (proven by a test against a different plant)", 1, False, None),
293 ], size=9.8, gap=4)
294 add_pic(sl, FIG / "robustness.png", 6.90, 2.45, 6.00, 4.35)
295
296 # ----- 5: Artifacts / Results
297 sl = S[4]
298 clear_body_shapes(sl)
299 set_title(sl, "ARTIFACTS – RESULTS", 26)
300 rows = [
301 ("A – Startup to target", 120, s["A"]["settled_Q"], s["A"]["settled_Q_std"],
302 s["A"]["final_choke"], s["A"]["min_bhp"]),
303 ("B – Target tracking 100→150", 150, s["B"]["settled_Q"], s["B"]["settled_Q_std"],
304 s["B"]["final_choke"], s["B"]["min_bhp"]),
305 ("C – Infeasible target 200", 200, s["C"]["settled_Q"], s["C"]["settled_Q_std"],
306 s["C"]["final_choke"], s["C"]["min_bhp"]),
307 ]
308 items = [("Scenario outcomes & tracking performance", 0, True, BLUE)]
309 for name, tgt, q, sd, u, bhp in rows:
310 verdict = "capped at max safe rate" if tgt == 200 else f"{abs(q - tgt) / tgt * 100:.1f} % error"
311 items.append((f"{name}: settled {q:.1f} ± {sd:.1f} bbl/hr (target {tgt}) . "
312 f"choke {u:.1f} % · min BHP {bhp:.0f} psi · {verdict}", 1, False, None))
313 items += [
314 ("", 0, False, None),
315 ("Safety performance", 0, True, BLUE),
316 (f"WHP ≥ {LIMITS.whp_min:.0f}, FLP ≤ {LIMITS.flp_max:.0f}, BHP ≥ {LIMITS.bhp_min:.0f} psi, "
317 f"|Δu| ≤ {CONTROLLER.max_move:.0f} %/step – held on EVERY sample", 1, False, None),
318 ("Automated PASS/FAIL audit in the pipeline; non-zero exit if any check fails", 1, False, None),
319 ("", 0, False, None),
320 ("Deliverables", 0, True, BLUE),
321 ("Simulator · step tests · model ID · MPC · 3 scenarios · Monte-Carlo · 19 tests · "
322 "dashboard · report", 1, False, None),
323 ("One command reproduces everything: python run_all.py", 1, True, INK),
324 ]
325 add_bullets(sl, 0.55, 1.22, 5.9, 3.05, items, size=9.2, gap=3)
326 add_pic(sl, FIG / "dashboard" / "dash_scenarioC_light.png", 6.55, 1.22, 6.30, 3.05)
327 add_pic(sl, FIG / "scenarios_compact.png", 0.55, 4.42, 12.30, 2.45)
328 add_bullets(sl, 0.55, 6.88, 12.30, 0.35, [
329 ("All three scenarios: target tracked when feasible (A, B); refused and capped at the safe "
330 "maximum when not (C) – BHP rests on its limit without ever crossing it.", 0, False, DIM),
331 ], size=9)
332
333 # ----- 6: Research & References
334 sl = S[5]
335 clear_body_shapes(sl)
336 set_title(sl, "RESEARCH AND REFERENCES", 26)
337 add_bullets(sl, 0.55, 1.30, 6.2, 5.3, [
338 ("Lessons learned", 0, True, BLUE),
339 ("Where you place step-test knots is a SAFETY decision – linear interpolation across a "
340 "convex curve is optimistic exactly where the constraint binds", 1, False, None),
341 ("A horizon shorter than the slowest time constant is not conservative, it is unsafe", 1, False, None),
342 ("One good run is not evidence; a safety claim about a stochastic system needs a distribution", 1, False,
None),
343 ("Margins must cover estimator lag and model error, not just sensor noise", 1, False, None),
344 ("Every filter is a trade: ours removed constraint hunting but cost response lag "
345 "that had to be paid for in margin", 1, False, None),

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346 ("Infeasibility needs no special-case code – it falls out of predict → filter → select", 1, False, None),
347 ("", 0, False, None),
348 ("Engineering references", 0, True, BLUE),
349 ("Inflow performance / linear IPR (constant productivity index) – standard above bubble point", 1, False,
None),
350 ("Choke orifice / valve-sizing relation  $Q = C_v \cdot f(u) \cdot \sqrt{\Delta P}$ ", 1, False, None),
351 ("Dynamic Matrix Control: move blocking, output disturbance (bias) estimation, "
352 "constraint back-off", 1, False, None),
353 ("Open-loop step testing and FOPDT identification for control-oriented models", 1, False, None),
354 ], size=11)
355 add_bullets(sl, 6.95, 1.30, 5.95, 5.3, [
356 ("Provided material used", 0, True, BLUE),
357 ("Autonomous Choke Control Simulated Dataset.csv – reference only, "
358 "as the problem statement directs; never used to fit the model", 1, False, None),
359 ("Compared against our physics stand-in and the differences reported openly "
360 "(reference is near-linear; its FLP falls with rate)", 1, False, None),
361 ("", 0, False, None),
362 ("Project artifacts", 0, True, BLUE),
363 ("src/ well_simulator · model · controller · simulator_interface", 1, False, None),
364 ("scripts/ reference EDA · step tests · model ID · scenarios · robustness · capture · PDF", 1, False, None),
365 ("tests/ 19 tests incl. a swap-in proof against a different plant", 1, False, None),
366 ("dashboard/ self-contained HTML/JS demo (no build, no server needed)", 1, False, None),
367 ("report/ENGINEERING_REPORT.md → submission/ENGINEERING_REPORT.pdf", 1, False, None),
368 ("", 0, False, None),
369 ("Note on the simulator", 0, True, RED),
370 ("The simulator module itself was not supplied, so we built a documented physics-based "
371 "stand-in from the process description. The controller touches only step()/reset(), "
372 "so the official simulator substitutes with zero changes.", 1, False, None),
373 ], size=11)
374
375 OUT.parent.mkdir(exist_ok=True)
376 prs.save(str(OUT))
377 print(f"wrote {OUT.relative_to(ROOT)} ({OUT.stat().st_size / 1024:.0f} KB, {len(S)} slides)")
378
379
380 if __name__ == "__main__":
381     main()
382

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